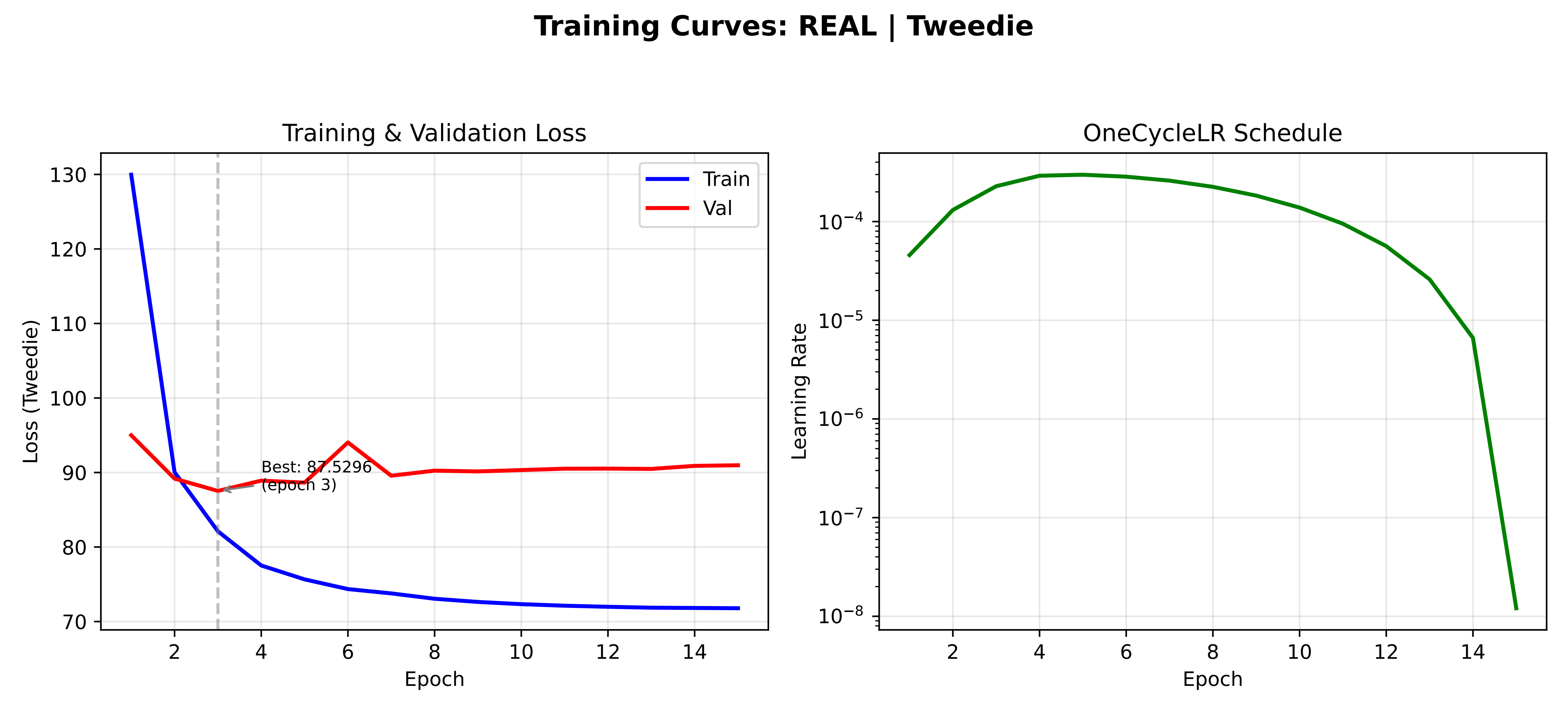
CrabTransformer Training Report

Data: REAL | Scenario: REAL | Loss: Tweedie

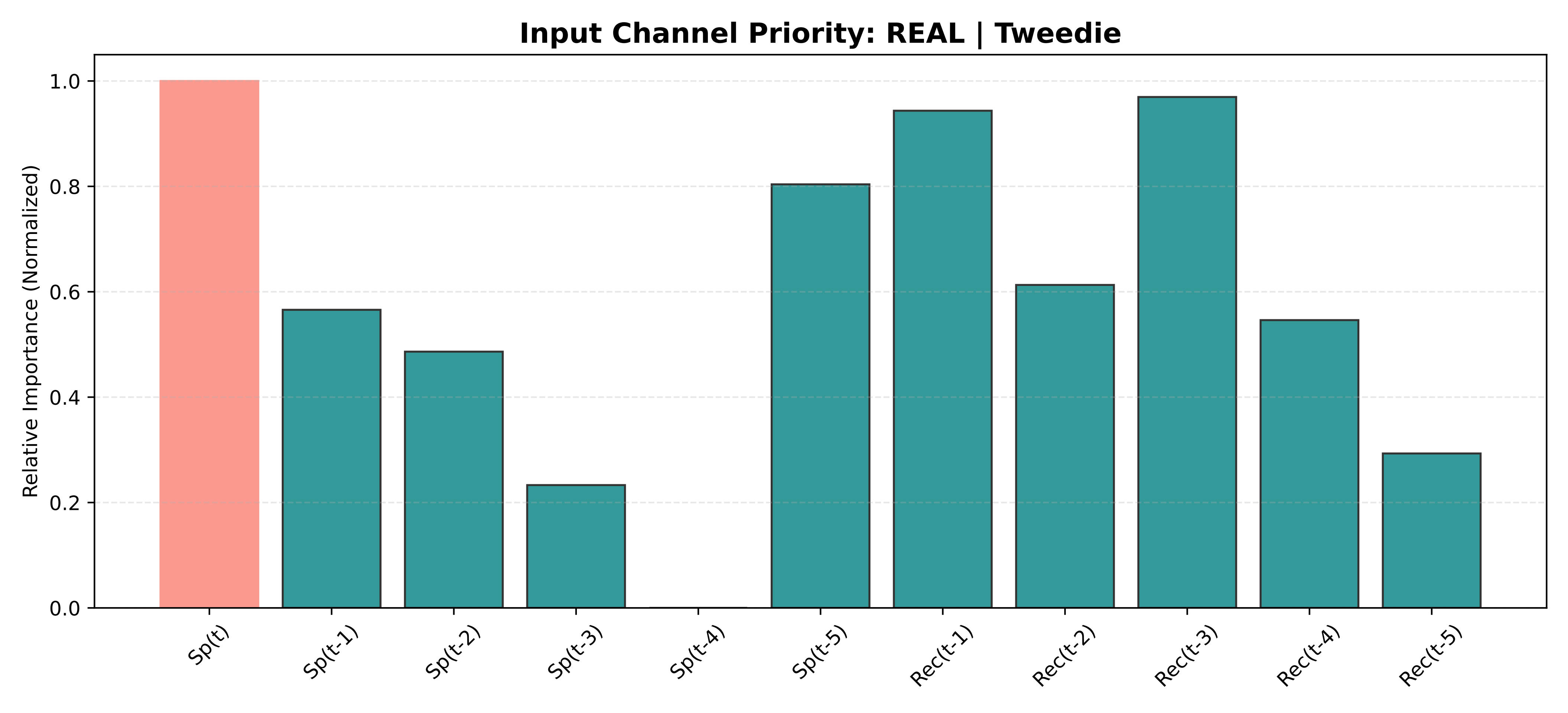
```
─── Real Data ───
  Source: Eastern Bering Sea Snow Crab Survey
  Grid: 50x50 (25km cells, SPDE interpolation)
  Valid ocean cells: 886
  Bootstraps: 100 (subsampling from ~349 stations)

─── Training Configuration ───
  epochs_run: 15     best_val_epoch: 3
  best_val_loss: 87.5296     final_train_loss: 71.7880
  bias_correction (MSE): 1.0000

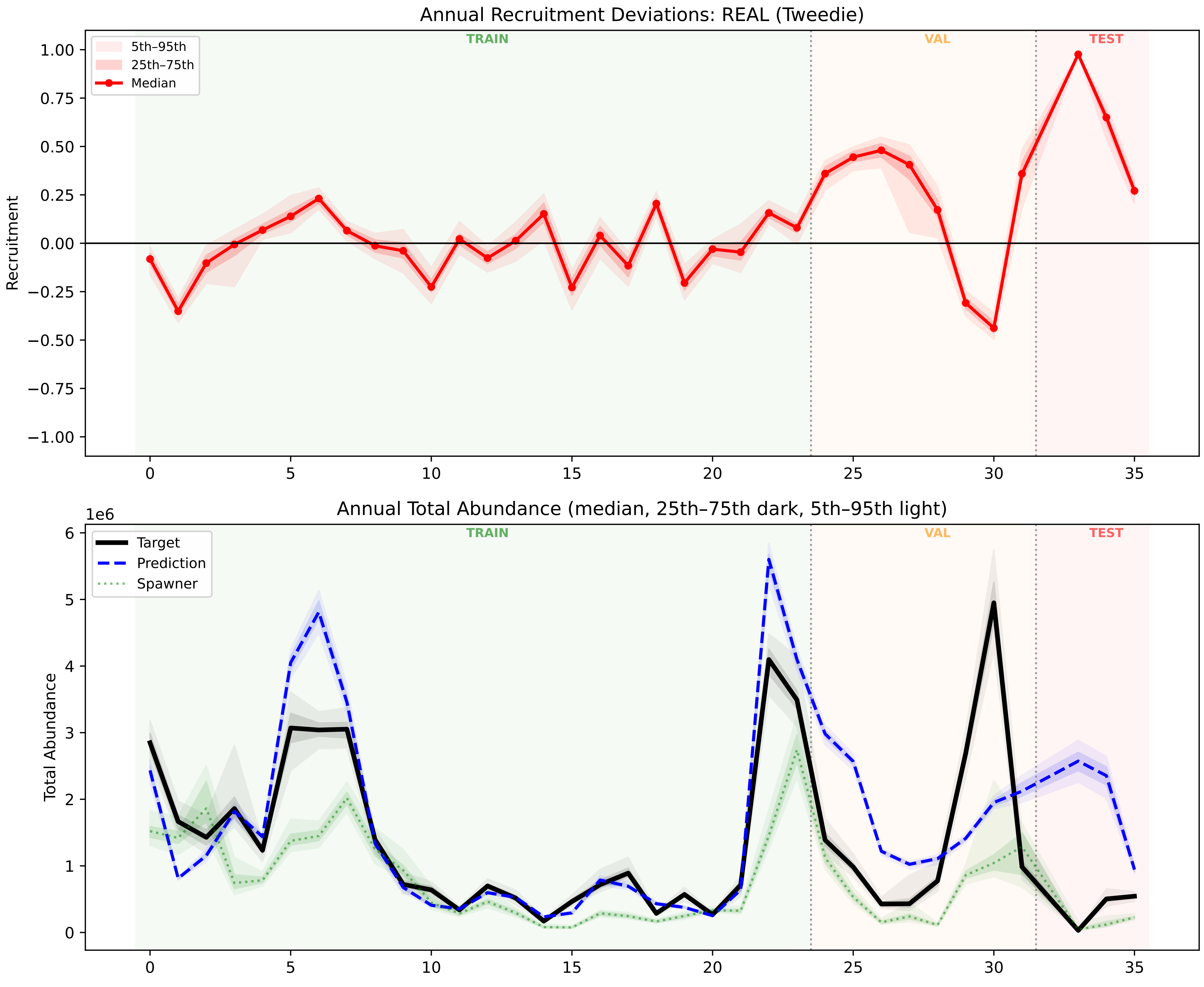
─── Model Architecture ───
  embed_dim: 64     num_heads: 4     num_layers: 3
  d_ff: 256     dropout: 0.2     patch_size: 5
  train/val/test years: 24/8/4
```



Training Curves: REAL | Tweedie
Training & Validation Loss
Train
Val
Best: 87.5296
(epoch 3)
Loss (Tweedie)
Epoch
OneCycleLR Schedule
Learning Rate
Epoch

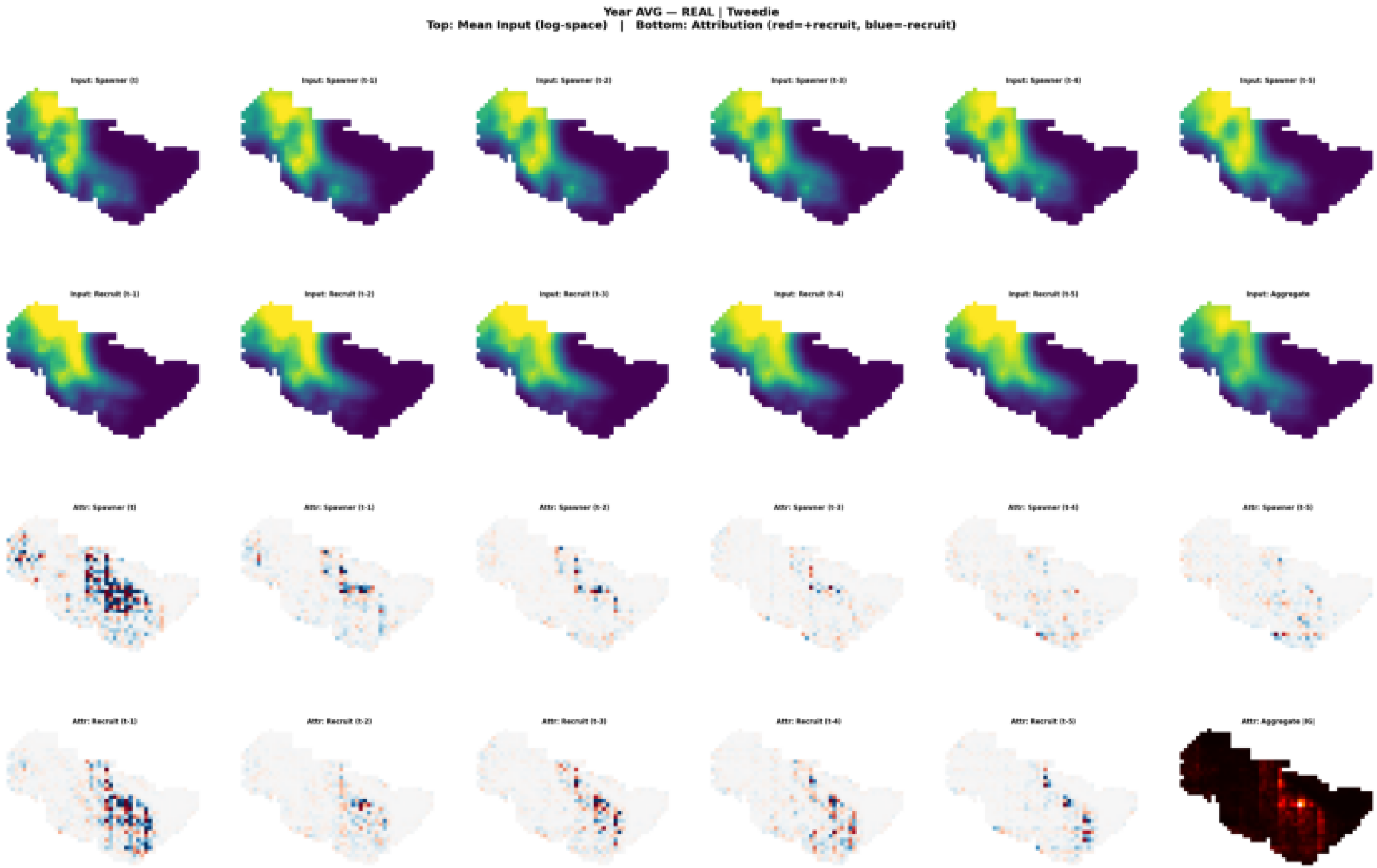


Input Channel Priority: REAL | Tweedie
Relative Importance (Normalized)
1.0
0.8
0.6
0.4
0.2
0.0
Sp(t)
Sp(t-1)
Sp(t-2)
Sp(t-3)
Sp(t-4)
Sp(t-5)
Rec(t-1)
Rec(t-2)
Rec(t-3)
Rec(t-4)
Rec(t-5)



Annual Recruitment Deviations: REAL (Tweedie)
TRAIN
VAL
TEST
5th–95th
25th–75th
Median
Recruitment
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
5
10
15
20
25
30
35
Annual Total Abundance (median, 25th–75th dark, 5th–95th light)
1e6
Target
Prediction
Spawner
Total Abundance
6
5
4
3
2
1
0

Spatial Trajectory: REAL | Tweedie


Year AVG — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
Input: Spawner (t-5)
Input: Recruit (t-1)
Input: Recruit (t-2)
Input: Recruit (t-3)
Input: Recruit (t-4)
Input: Recruit (t-5)
Input: Aggregate
Attr: Spawner (t)
Attr: Spawner (t-1)
Attr: Spawner (t-2)
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Attr: Spawner (t-4)
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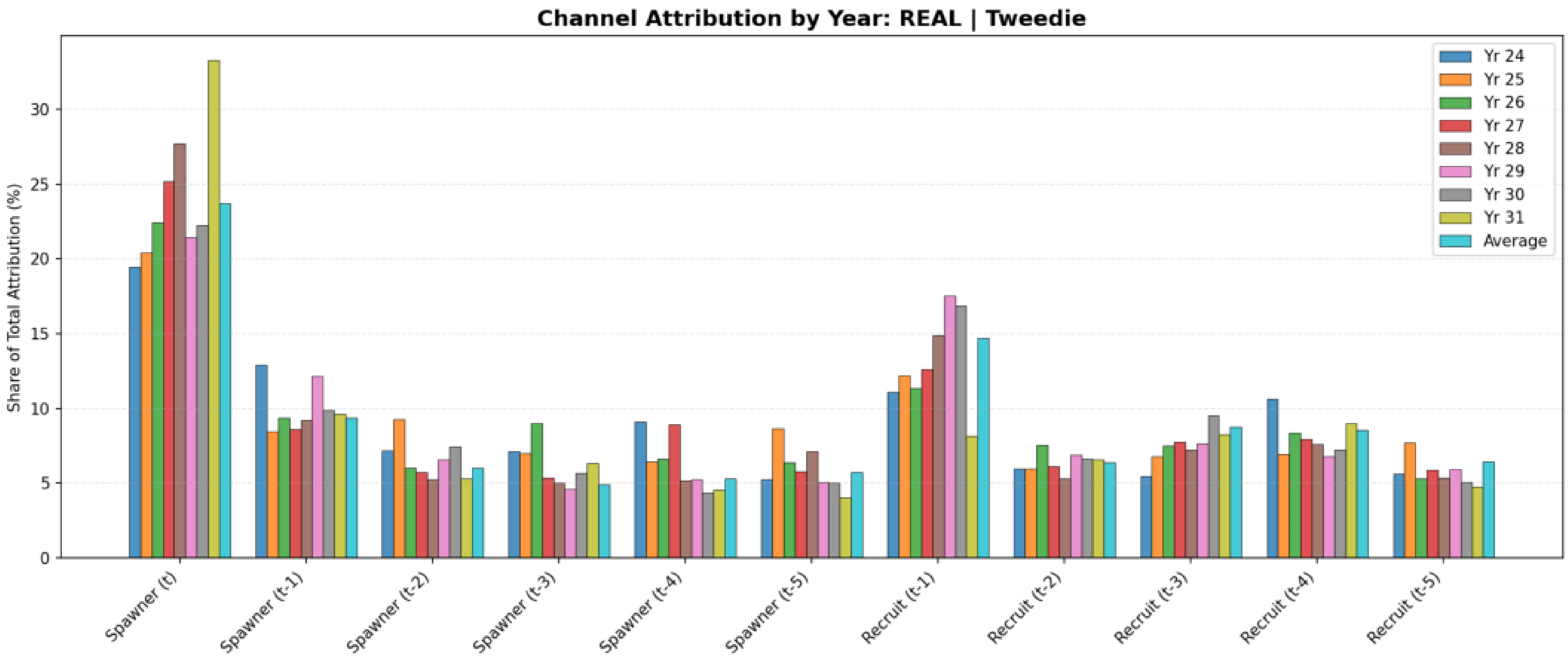
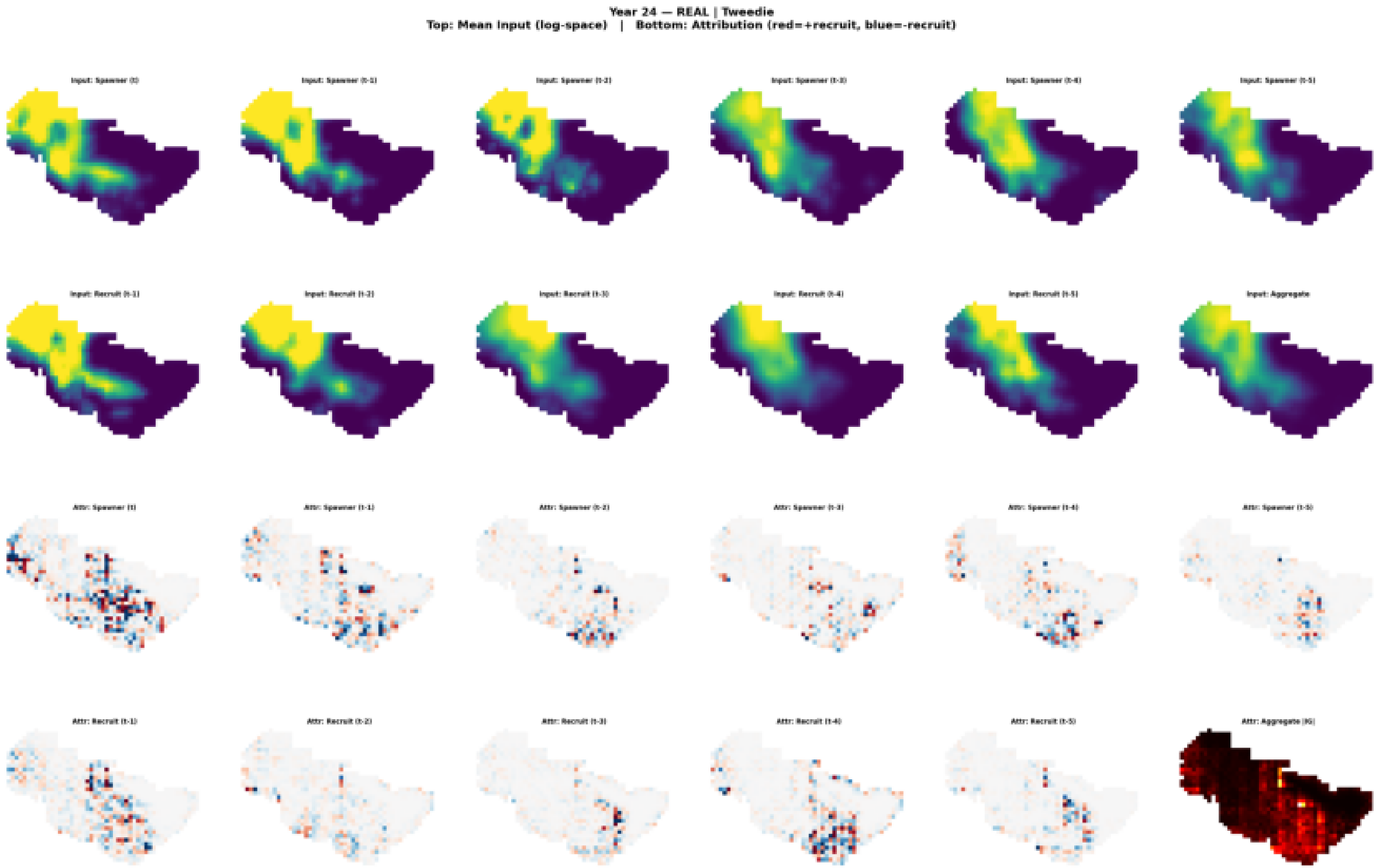
Attr: Recruit (t-4)
Attr: Recruit (t-5)
Attr: Aggregate (IG)



Channel Attribution by Year: REAL | Tweedie
Share of Total Attribution (%)
0
5
10
15
20
25
30
Yr 24
Yr 25
Yr 26
Yr 27
Yr 28
Yr 29
Yr 30
Yr 31
Average
Spawner (t)
Spawner (t-1)
Spawner (t-2)
Spawner (t-3)
Spawner (t-4)
Spawner (t-5)
Recruit (t-1)
Recruit (t-2)
Recruit (t-3)
Recruit (t-4)
Recruit (t-5)


Year 24 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
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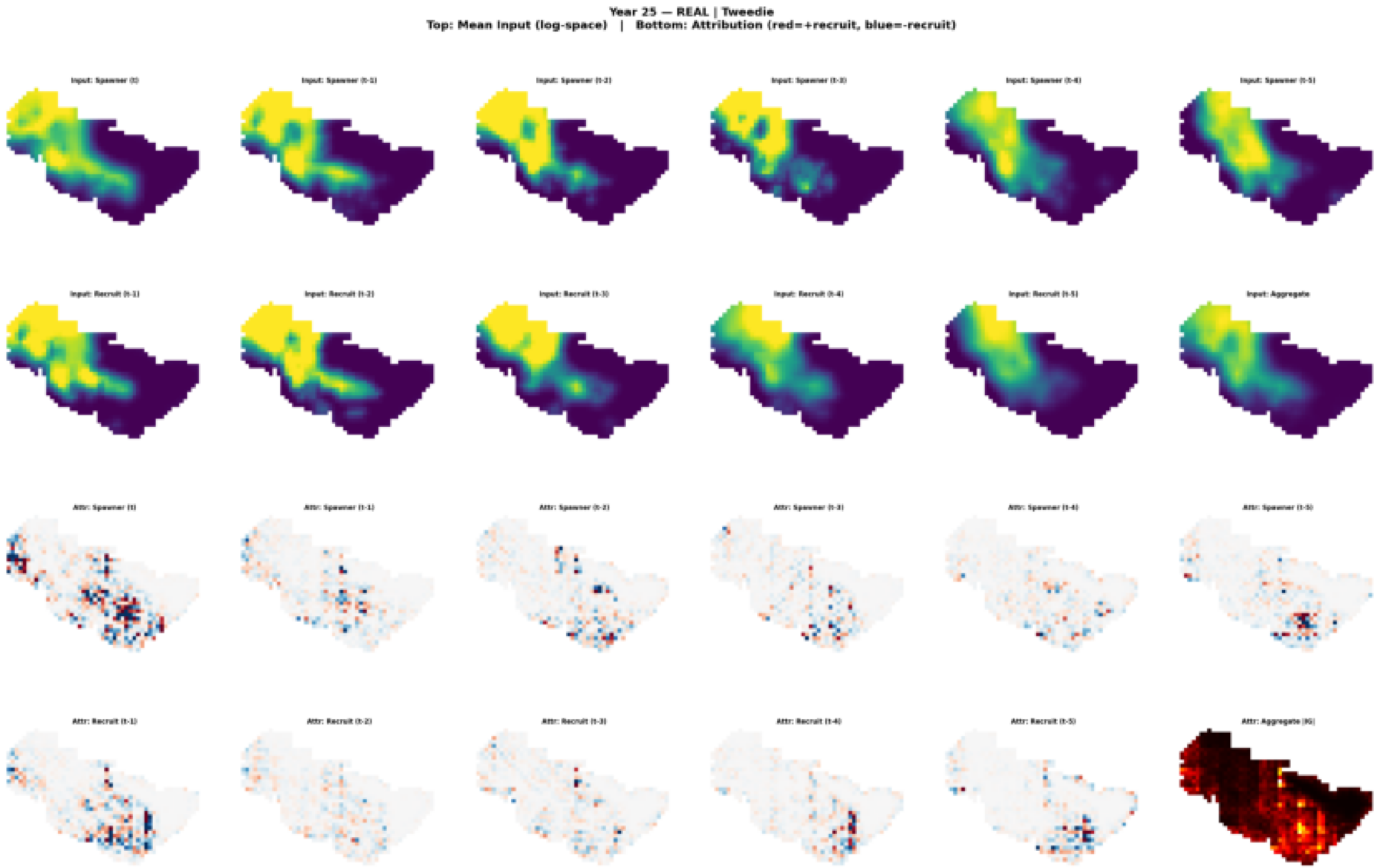
Year 25 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
Input: Spawner (t-5)
Input: Recruit (t-1)
Input: Recruit (t-2)
Input: Recruit (t-3)
Input: Recruit (t-4)
Input: Recruit (t-5)
Input: Aggregate
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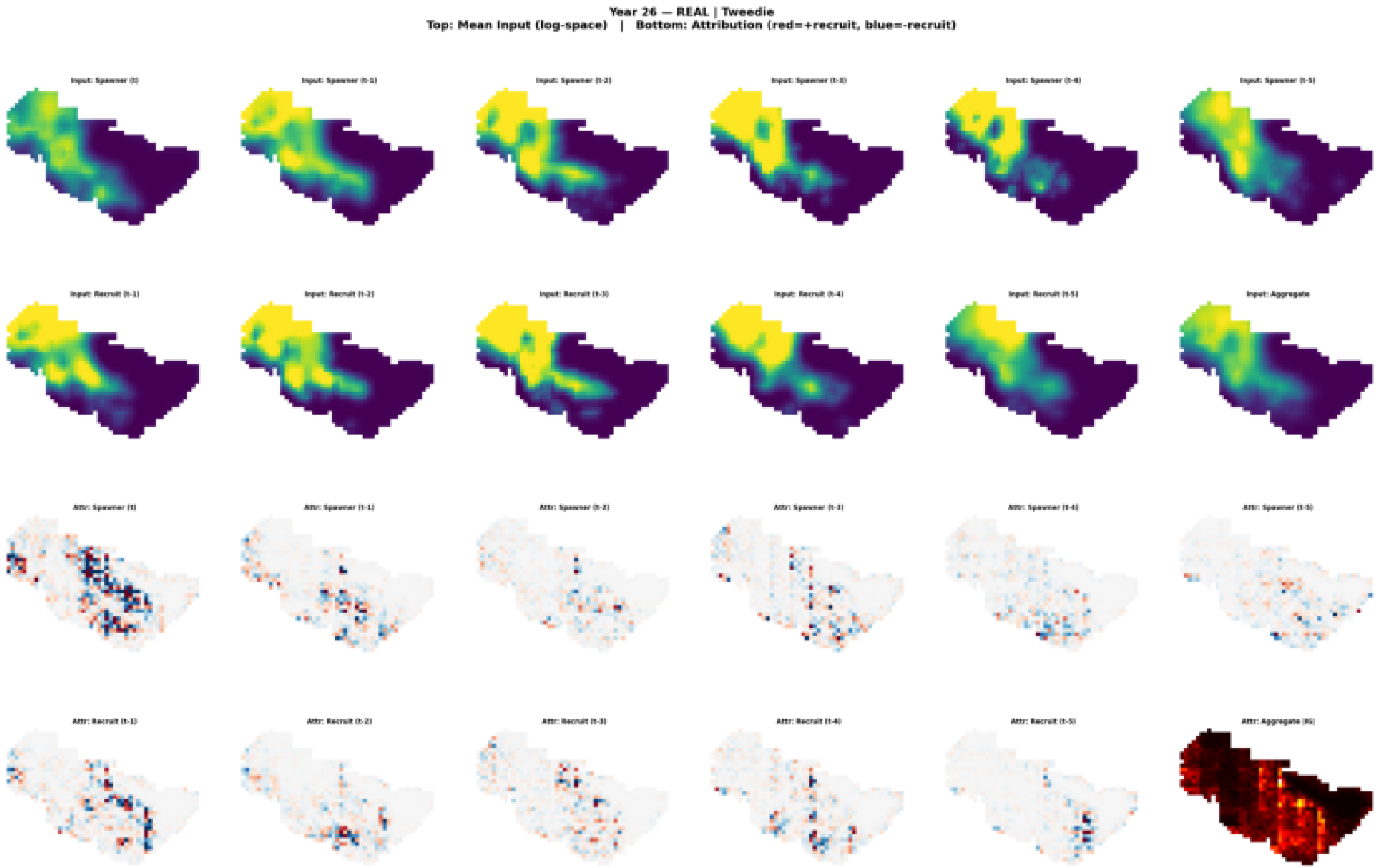
Attr: Spawner (t-1)
Attr: Spawner (t-2)
Attr: Spawner (t-3)
Attr: Spawner (t-4)
Attr: Spawner (t-5)
Attr: Recruit (t-1)
Attr: Recruit (t-2)
Attr: Recruit (t-3)
Attr: Recruit (t-4)
Attr: Recruit (t-5)
Attr: Aggregate (IG)


Year 26 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
Input: Spawner (t-5)
Input: Recruit (t-1)
Input: Recruit (t-2)
Input: Recruit (t-3)
Input: Recruit (t-4)
Input: Recruit (t-5)
Input: Aggregate
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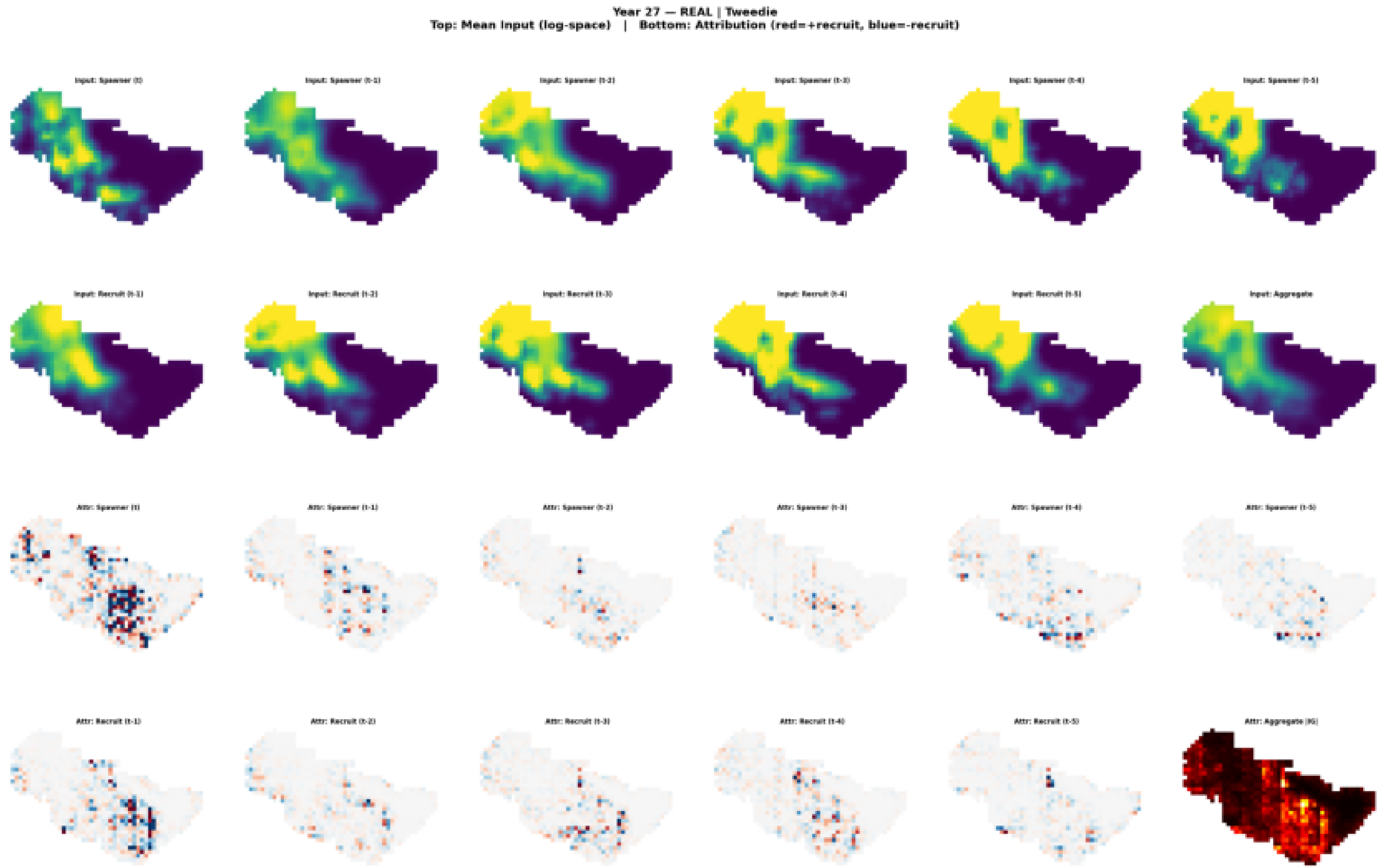
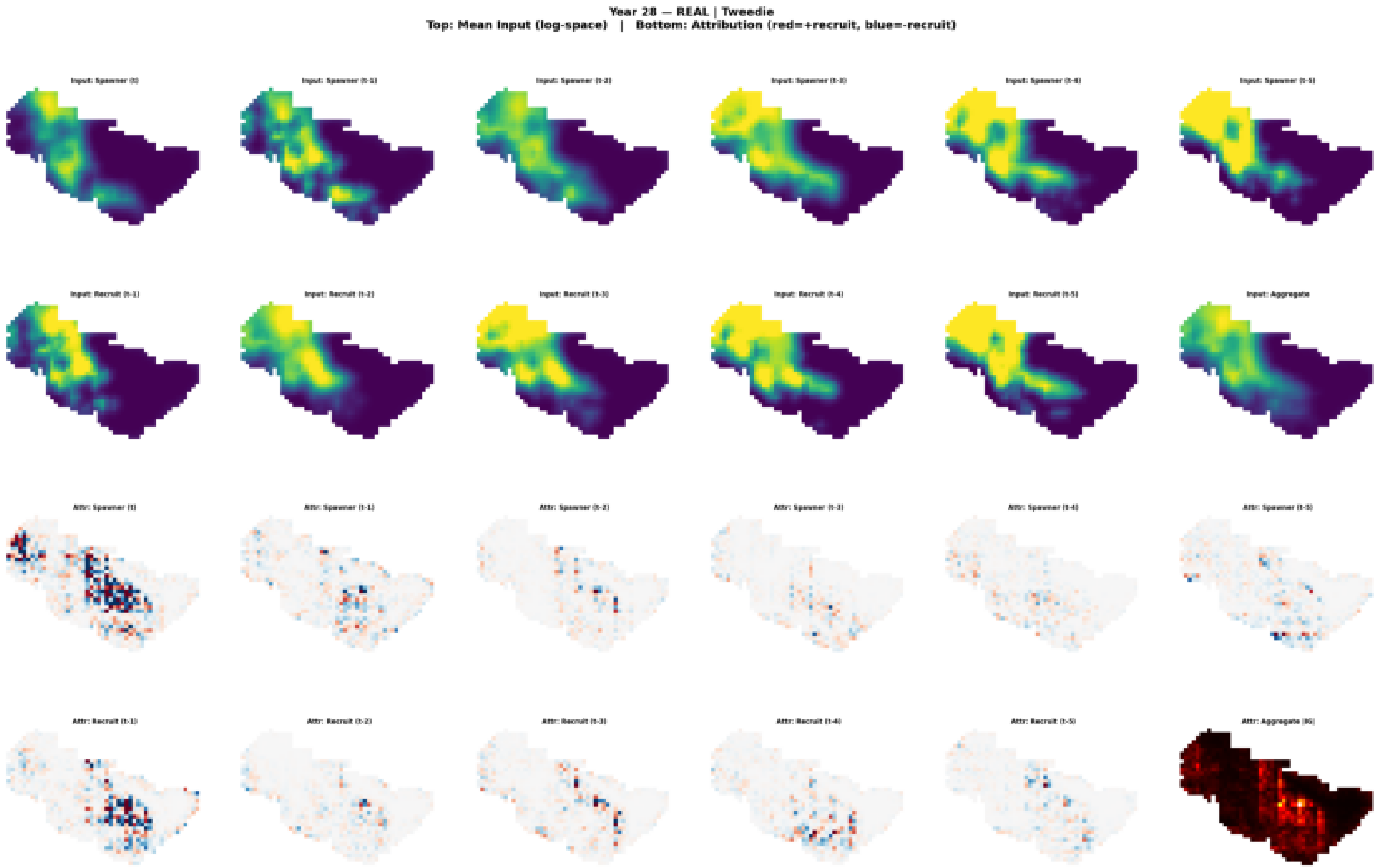
Attr: Recruit (t-4)
Attr: Recruit (t-5)
Attr: Aggregate (IG)


Year 27 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 28 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
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Input: Recruit (t-1)
Input: Recruit (t-2)
Input: Recruit (t-3)
Input: Recruit (t-4)
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Input: Aggregate
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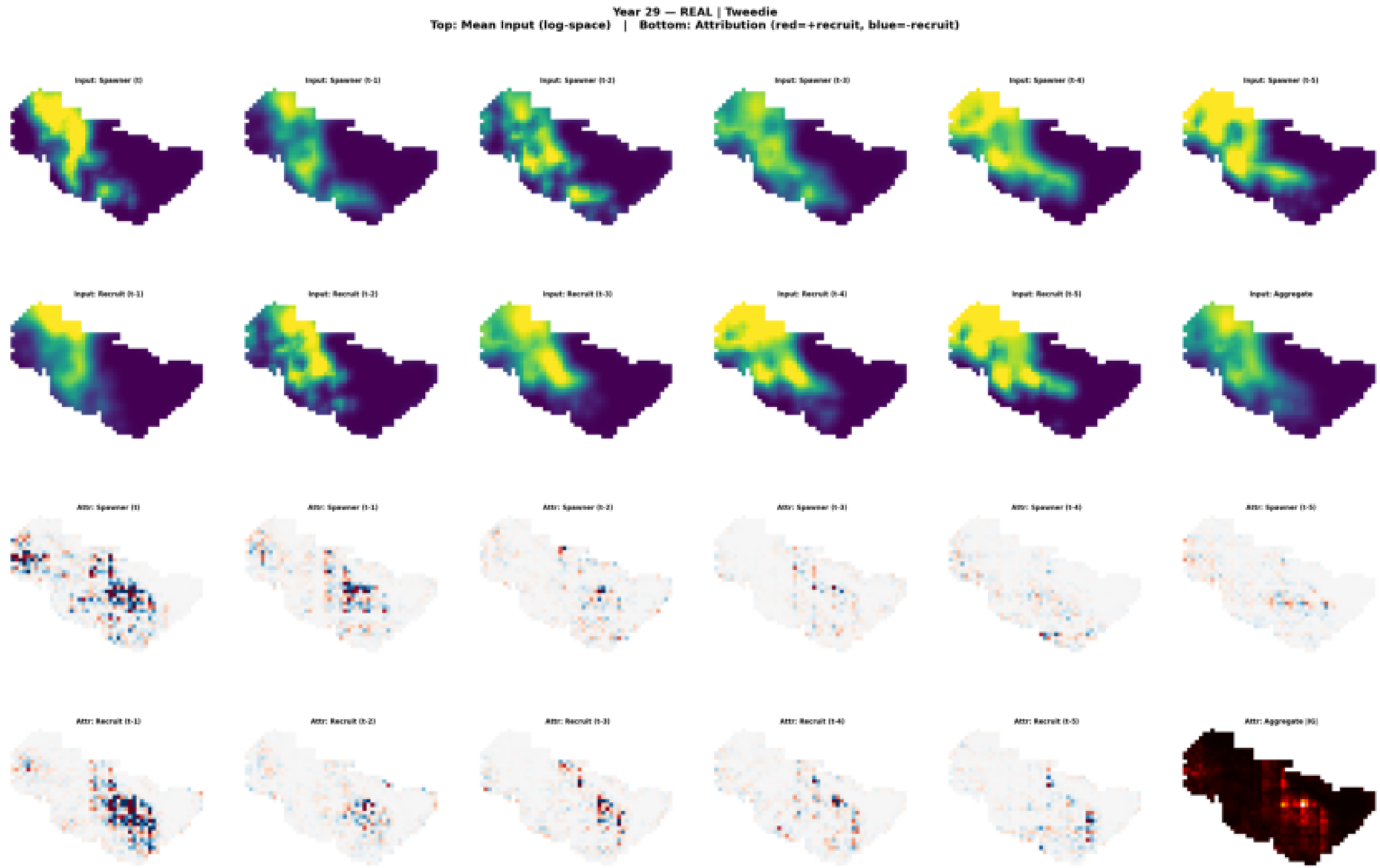
Attr: Spawner (t-5)
Attr: Recruit (t-1)
Attr: Recruit (t-2)
Attr: Recruit (t-3)
Attr: Recruit (t-4)
Attr: Recruit (t-5)
Attr: Aggregate (IG)


Year 29 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
Input: Spawner (t-5)
Input: Recruit (t-1)
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Attr: Spawner (t)
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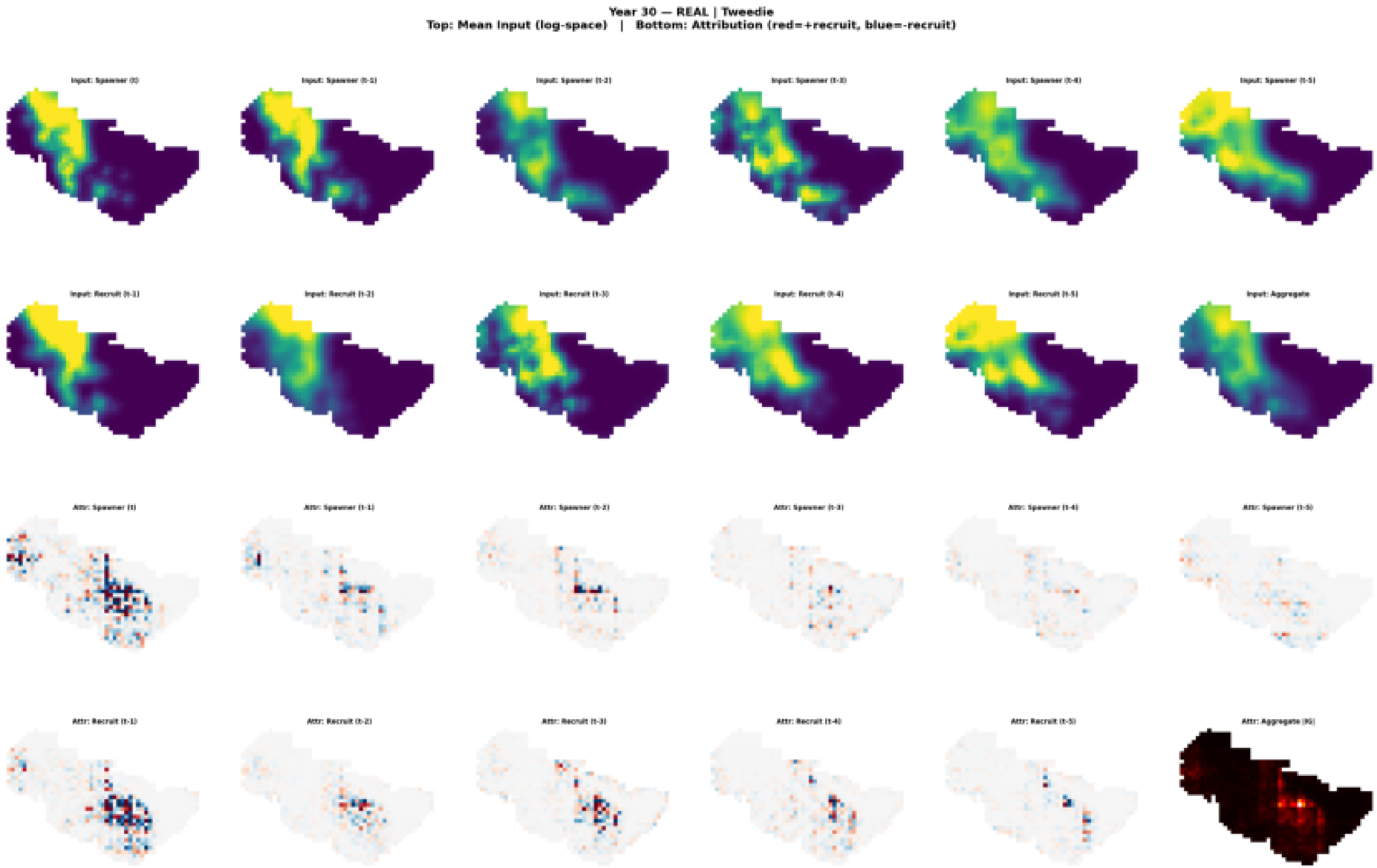

Year 30 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
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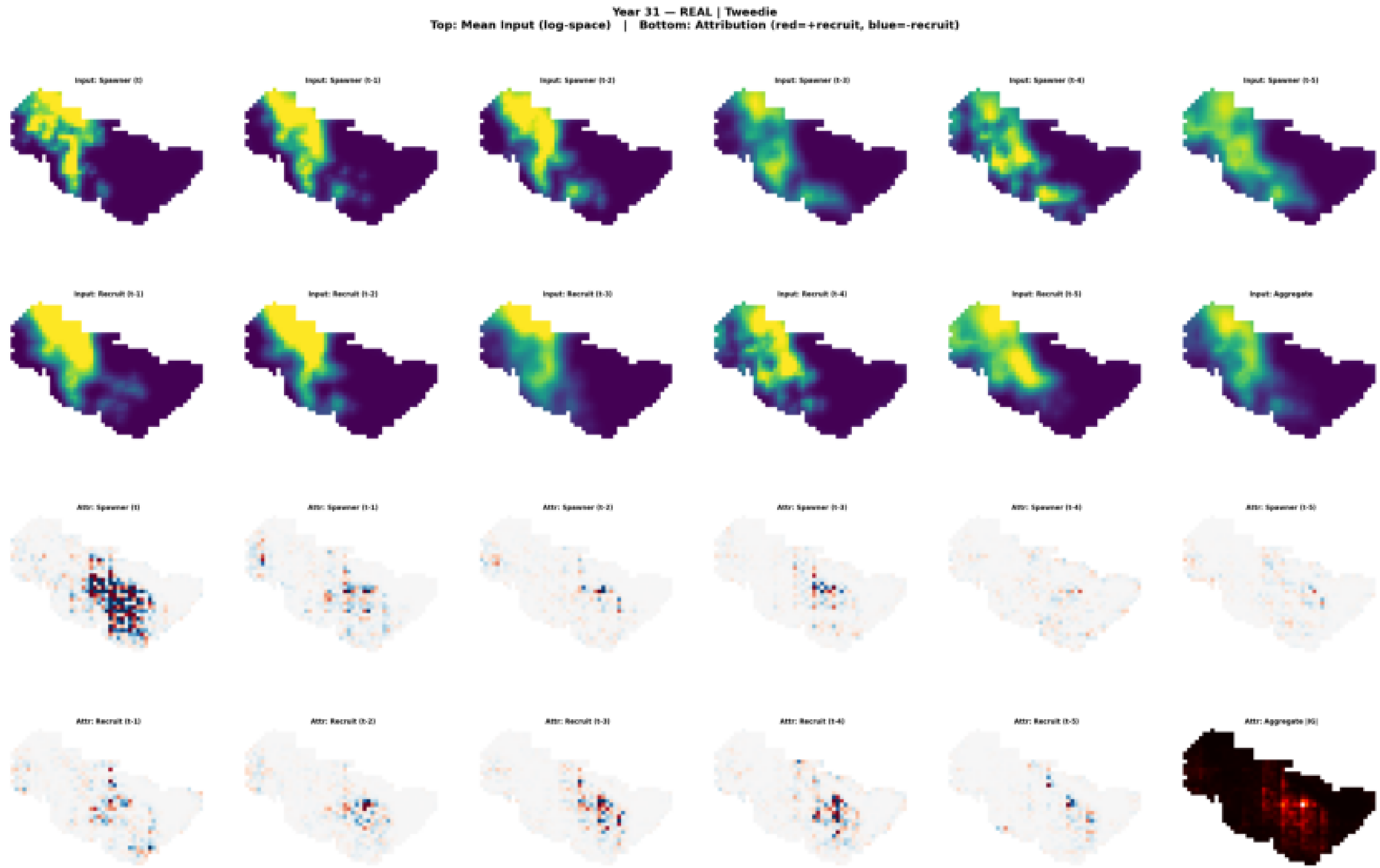
Year 31 — REAL | Tweedie
Top: Mean Input (log-space) | Bottom: Attribution (red=+recruit, blue=-recruit)
Input: Spawner (t)
Input: Spawner (t-1)
Input: Spawner (t-2)
Input: Spawner (t-3)
Input: Spawner (t-4)
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Attr: Spawner (t-4)
Attr: Spawner (t-5)
Attr: Recruit (t-1)
Attr: Recruit (t-2)
Attr: Recruit (t-3)
Attr: Recruit (t-4)
Attr: Recruit (t-5)
Attr: Aggregate (IG)

Summary Statistics: REAL | Tweedie

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance	MAE	Spearman
TRAIN	1750.5	1612.8	6719.9	0.0	8622.4	1.085	851.4	0.911
VAL	2027.6	1801.5	5039.4	0.0	14523.2	1.126	1874.0	0.853
TEST	2206.4	406.3	7110.9	0.0	16679.4	5.430	1844.9	0.794

Detailed Metrics (R-style): REAL | Tweedie

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	2381.7	2337.5	137.6	2.7%	0.918	0.907	0.1%	0.903	0.828	0.710	0.66	0.721	0.684
VAL	5582.6	5382.4	226.1	7.8%	0.855	0.713	75.4%	0.869	0.690	0.543	0.99	-0.990	0.133
TEST	5832.1	5546.3	1800.0	8.8%	0.800	0.699	2964.9%	0.851	0.784	0.657	1.14	-3104.242	-18.123

Appendix A: Summary Metrics

APPENDIX A: Table Metric Definitions

Data type: REAL
Split: 24 train / 8 val / 4 test years

Spatial masking: All metrics computed ONLY over valid ocean cells (~886) in the 50x50 grid. Padding cells are excluded from all computations.

All metrics computed on ORIGINAL DENSITY SCALE (after expm1 and bias correction for MSE). Zero threshold = 14 (smallest nonzero observed value): predictions AND observations below 14 are set to 0 before ALL metric computation. Both Page 7 and Page 8 apply this threshold.

Summary Table (Page 7)

Computed per bootstrap sample, then averaged across all samples in phase.

Pred_Mean, Tgt_Mean: Arithmetic mean of valid cell values.
Pred_SD: Standard deviation of valid cell values.
Pred_P50, Pred_P95: Median and 95th percentile.
Abundance: sum(pred) / sum(target). 1.0 = perfect.
MAE: mean(|pred - target|) per valid cell, averaged across samples.
Spearman: Rank correlation over valid cells, averaged across samples.

Appendix A (cont): Detailed Metrics

APPENDIX A (cont): Detailed Metrics (Page 8)

Per-year: average spatial field across bootstraps, then compute metrics on that mean field over valid cells. Average across years per phase.

RMSE: sqrt(mean((obs - pred)^2)) over valid cells.
RMSE_unbiased: RMSE after subtracting mean bias.
Bias: mean(pred - obs). Positive = overprediction.
Bias_%: bias^2 / rmse^2 * 100.
Spearman / Pearson: Rank and linear correlation.
Pct_Error: (sum(pred) - sum(obs)) / sum(obs) * 100.
Zero_F1: F1 for zero-classification (threshold = 14).
TopK_Accuracy / Jaccard: Overlap of top-10% density cells.
Quantile_Mean_Error: Mean |bin diff| across 10 quantile bins.
Skill_MSE / Skill_MAE: 1 - error/climatology. Positive = beats mean.

Appendix B: Plot Methodology

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APPENDIX B: Plot Methodology
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Page 2: Training Curves
  MSE loss in log-space; Tweedie on original scale.
  OneCycleLR: max_lr/10 -> max_lr (10%), cosine to max_lr/1000.

Page 3: Channel Importance
  mean(|weights|) of each channel's projection. Static, not causal.

Page 4: RecDev & Abundance
  RecDev = (pred - target) / (target + pred). Sums over valid cells.
  Ribbons: 5th-95th and 25th-75th percentiles across bootstraps.

Page 5: Spatial Trajectory
  Log-space grids, first bootstrap, shared colorscale 0-8.

Page 6: Integrated Gradients
  Pre-computed, averaged across bootstraps per year.
  Output function sums over valid cells only (real data).

Bias Correction (MSE only)
  pred = expm1(pred_log) * exp(sigma^2 / 2)
  Tweedie: no correction needed.
```